

GATE Data Science & Artificial Intelligence (DA)

Comprehensive 80-Question Topic-Wise Question Bank

10 PYQs + 10 Practice Questions with Detailed Step-by-Step Solutions per Topic

Topics: Diagonalization & SVD, Bias-Variance & Overfitting, Indexing & Hashing, Verbal Ability

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1 Topic 1: Diagonalization, Symmetric/Orthogonal Matrices, SVD Intro

1.1 10 Previous Year Questions (PYQs)

Q1.1

[GATE DA 2024 | NAT | 2 Marks]

Let $u = [1, 2, 3, 4, 5]^T$, and let $\sigma_1, \sigma_2, \sigma_3, \sigma_4, \sigma_5$ be the singular values of the matrix $M = uu^T$ (where u^T is the transpose of u). The value of $\sum_{i=1}^5 \sigma_i$ is _____. **Answer: 55** **Solution:** $M = uu^T$ is an outer product of non-zero vector u , so $\text{rank}(M) = 1$. A rank-1 matrix has exactly 1 non-zero singular value, while $\sigma_2 = \sigma_3 = \sigma_4 = \sigma_5 = 0$. Since M is symmetric and positive semi-definite, its non-zero singular value equals its non-zero eigenvalue: $Mu = (uu^T)u = u(u^Tu) = \|u\|_2^2 u$. Thus $\sigma_1 = \|u\|_2^2 = 1^2 + 2^2 + 3^2 + 4^2 + 5^2 = 1 + 4 + 9 + 16 + 25 = 55$. The sum of singular values is $55 + 0 + 0 + 0 + 0 = 55$.

Q1.2

[GATE DA 2025 | MSQ | 2 Marks]

Let A be a real symmetric matrix of order $n \times n$. Which of the following statements is/are ALWAYS true?

- (A) There exists an orthogonal matrix Q such that $Q^T A Q$ is a diagonal matrix.
- (B) All eigenvalues of A are real numbers.
- (C) If $\lambda_1 \neq \lambda_2$ are distinct eigenvalues of A with eigenvectors v_1 and v_2 , then $v_1^T v_2 = 0$.
- (D) The singular values of A are the absolute values of the eigenvalues of A .

Answer: (A, B, C, D) **Solution:** All four statements are direct consequences of the Spectral Theorem for real symmetric matrices. (A) Orthogonal diagonalizability. (B) Eigenvalues are real. (C) Distinct eigenspaces are mutually orthogonal. (D) For symmetric A , $A^T A = A^2$, so singular values $\sigma_i = \sqrt{\lambda_i(A^2)} = \sqrt{\lambda_i^2} = |\lambda_i|$.

Q1.3

[GATE CS 2023 | MCQ | 1 Mark]

An $n \times n$ matrix A is diagonalizable if and only if:

- (A) It has n linearly independent eigenvectors.
- (B) All of its eigenvalues are strictly positive.
- (C) $\det(A) \neq 0$.
- (D) It is an upper triangular matrix.

Answer: (A) **Solution:** By the Fundamental Diagonalization Theorem, A is similar to a diagonal matrix ($A = PDP^{-1}$) if and only if A possesses n linearly independent eigenvectors, which form the columns of the invertible matrix P .

Q1.4

[GATE DA 2024 | MCQ | 1 Mark]

Let A be an $m \times n$ real matrix with Singular Value Decomposition $A = U\Sigma V^T$. Which of the following statements is FALSE?

- (A) The columns of U are the orthonormal eigenvectors of AA^T .
- (B) The columns of V are the orthonormal eigenvectors of $A^T A$.
- (C) The non-zero singular values of A are the square roots of the non-zero eigenvalues of $A^T A$.
- (D) Σ is always a square matrix of size $\min(m, n) \times \min(m, n)$.

Answer: (D) Solution: In the standard full SVD, U is $m \times m$, V is $n \times n$, and Σ is of dimension $m \times n$ matching the dimensions of A . Hence, Σ is not necessarily a square matrix.

Q1.5

[GATE CS 2021 | NAT | 2 Marks]

Let $Q = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ be a 2×2 orthogonal rotation matrix. What is the value of $\det(Q^T Q)$?

Answer: 1 Solution: For any orthogonal matrix Q , $Q^T Q = I$. Thus $\det(Q^T Q) = \det(I) = 1$.

Q1.6

[GATE DA 2026 | NAT | 2 Marks]

Let $A = \begin{bmatrix} 3 & 0 \\ 0 & -4 \end{bmatrix}$. What is the largest singular value of A ? **Answer: 4 Solution:** The singular

values of a diagonal matrix are the absolute values of its diagonal entries. Here, $\sigma_1 = |-4| = 4$ and $\sigma_2 = |3| = 3$. The largest singular value is 4.

Q1.7

[GATE CS 2020 | MCQ | 1 Mark]

If A is an orthogonal matrix ($A^T A = I$), which of the following is NOT necessarily true?

- (A) $\|Ax\|_2 = \|x\|_2$ for any vector x .
- (B) $\det(A) = 1$ or $\det(A) = -1$.
- (C) $A^{-1} = A^T$.
- (D) All eigenvalues of A must be strictly real.

Answer: (D) Solution: Orthogonal matrices preserve lengths (isometries) and have $\det = \pm 1$, but their eigenvalues can be complex numbers on the unit circle (e.g., a 90° 2D rotation matrix has eigenvalues $\pm i$).

Q1.8

[GATE DA 2025 | NAT | 2 Marks]

Let A be a 3×3 matrix with singular values $\sigma_1 = 5, \sigma_2 = 3, \sigma_3 = 0$. What is the Frobenius norm $\|A\|_F$? (Round off to two decimal places) **Answer: 5.83 Solution:** The Frobenius norm equals the square root of the sum of squared singular values: $\|A\|_F = \sqrt{\sum \sigma_i^2} = \sqrt{5^2 + 3^2 + 0^2} = \sqrt{25 + 9} = \sqrt{34} \approx 5.83$.

Q1.9

[GATE CS 2022 | MCQ | 2 Marks]

Let A be a 2×2 non-diagonal matrix with eigenvalues $\lambda_1 = 1$ and $\lambda_2 = 1$. The geometric multiplicity of $\lambda = 1$ is 1. Which of the following is TRUE?

- (A) A is defective and cannot be diagonalized.
- (B) A is symmetric.
- (C) A is orthogonal.
- (D) A has 2 linearly independent eigenvectors.

Answer: (A) Solution: A matrix is diagonalizable if and only if the geometric multiplicity equals the algebraic multiplicity for every eigenvalue. Here, algebraic multiplicity is 2 while geometric multiplicity is 1 ($1 < 2$). Thus, A is defective and not diagonalizable.

Q1.10

[GATE CS 2019 | NAT | 1 Mark]

Let $u = [1, 0]^T$ and $v = [0, 1]^T$. What is the rank of the matrix $M = 3uv^T$? **Answer: 1 Solution:** The outer product of two non-zero vectors uv^T always produces a rank-1 matrix.

1.2 10 Practice Questions (GATE Level)

P1.1

[Practice | NAT | 2 Marks]

Let $A = \begin{bmatrix} 0 & 3 \\ 4 & 0 \end{bmatrix}$. What is the largest singular value of A ? **Answer: 4** **Solution:** Compute $A^T A = \begin{bmatrix} 0 & 4 \\ 3 & 0 \end{bmatrix} \begin{bmatrix} 0 & 3 \\ 4 & 0 \end{bmatrix} = \begin{bmatrix} 16 & 0 \\ 0 & 9 \end{bmatrix}$. The eigenvalues of $A^T A$ are 16 and 9. The singular values are $\sigma_1 = \sqrt{16} = 4$ and $\sigma_2 = \sqrt{9} = 3$. The largest singular value is 4.

P1.2

[Practice | MCQ | 1 Mark]

The Moore-Penrose pseudoinverse A^+ of a matrix A with SVD $A = U\Sigma V^T$ is computed as:

(A) $A^+ = V\Sigma^+U^T$

(B) $A^+ = U^T\Sigma^+V$

(C) $A^+ = V^T\Sigma^+U$

(D) $A^+ = U\Sigma^+V^T$

Answer: (A) **Solution:** $A^+ = (U\Sigma V^T)^+ = (V^T)^+\Sigma^+U^+ = V\Sigma^+U^T$, where Σ^+ is obtained by taking the reciprocals of non-zero singular values and transposing.

P1.3

[Practice | MSQ | 2 Marks]

Which of the following statements about matrix diagonalization is/are TRUE?

(A) If an $n \times n$ matrix has n distinct eigenvalues, it is always diagonalizable.

(B) Every real symmetric matrix is diagonalizable.

(C) If a matrix is invertible, it is always diagonalizable.

(D) If $A = PDP^{-1}$, then $A^k = PD^kP^{-1}$ for any positive integer k .

Answer: (A, B, D) **Solution:** (A) Distinct eigenvalues guarantee linearly independent eigenvectors. (B) Spectral Theorem. (C) is false: e.g., $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ is invertible ($\det = 1$) but not diagonalizable. (D) Standard power identity.

P1.4

[Practice | NAT | 2 Marks]

Let A be a 3×2 matrix whose non-zero singular values are $\sigma_1 = 6$ and $\sigma_2 = 2$. What is the condition number $\kappa(A) = \frac{\sigma_{\max}}{\sigma_{\min}}$? **Answer: 3** **Solution:** $\kappa(A) = \frac{\sigma_1}{\sigma_2} = \frac{6}{2} = 3$.

P1.5

[Practice | MCQ | 2 Marks]

By the Eckart-Young-Mirsky Theorem, the best rank- k approximation A_k of a matrix A in terms of Frobenius norm is obtained by:

(A) Truncating the SVD of A to retain only the top k largest singular values: $\sum_{i=1}^k \sigma_i u_i v_i^T$.

(B) Setting the last k rows of A to zero.

(C) Applying Gram-Schmidt orthogonalization to the first k columns.

(D) Computing the QR factorization and keeping the top k diagonal elements.

Answer: (A) **Solution:** The Eckart-Young theorem states that the optimal rank- k approximation under both spectral and Frobenius norms is obtained by keeping the k largest singular values and corresponding singular vectors in the SVD expansion.

P1.6

[Practice | NAT | 1 Mark]

If Q is an orthogonal matrix, what is the value of $\|Qx\|_2$ when $\|x\|_2 = 7$? **Answer: 7** **Solution:** Orthogonal matrices preserve Euclidean norms: $\|Qx\|_2^2 = (Qx)^T(Qx) = x^T Q^T Q x = x^T I x = \|x\|_2^2$. Therefore, $\|Qx\|_2 = \|x\|_2 = 7$.

P1.7

[Practice | MSQ | 2 Marks]

Let $A = U\Sigma V^T$ be the SVD of an $m \times n$ matrix with rank r . Which of the following spaces is/are correctly identified?

- (A) The first r columns of U form an orthonormal basis for the column space of A .
- (B) The last $m - r$ columns of U form an orthonormal basis for the left null space of A .
- (C) The first r columns of V form an orthonormal basis for the row space of A .
- (D) The last $n - r$ columns of V form an orthonormal basis for the null space of A .

Answer: (A, B, C, D) **Solution:** These represent the Four Fundamental Subspaces characterized directly by the singular vectors in SVD.

P1.8

[Practice | NAT | 2 Marks]

Let $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$. The eigenvalues of A are $\lambda_1 = 3$ and $\lambda_2 = 1$. What is the 2-norm (spectral norm) of A , defined as $\|A\|_2 = \sigma_{\max}(A)$? **Answer: 3** **Solution:** For a real symmetric positive semi-definite matrix, the singular values equal the eigenvalues. Thus $\sigma_1 = 3$ and $\sigma_2 = 1$. The spectral norm is $\sigma_{\max} = 3$.

P1.9

[Practice | MCQ | 1 Mark]

If A and B are similar matrices ($B = P^{-1}AP$), which of the following properties is NOT necessarily shared by A and B ?

- (A) Eigenvalues
- (B) Determinant
- (C) Trace
- (D) Eigenvectors

Answer: (D) **Solution:** Similar matrices have identical characteristic polynomials, eigenvalues, determinants, and traces. However, their eigenvectors differ: if v is an eigenvector of A , then $P^{-1}v$ is an eigenvector of B .

P1.10

[Practice | NAT | 1 Mark]

A 4×4 real matrix has singular values 4, 3, 2, 0. What is the rank of this matrix? **Answer: 3** **Solution:** The rank of a matrix equals the number of its non-zero singular values. Here there are 3 non-zero singular values (4, 3, 2), so the rank is 3.

2 Topic 2: Bias-Variance Tradeoff, Overfitting/Underfitting Deep-Dive

2.1 10 Previous Year Questions (PYQs)

Q2.1

[GATE DA 2024 | MCQ | 1 Mark]

In supervised learning, the expected mean squared test error of a regression model at a query point x can be decomposed into:

- (A) $\text{Bias}^2(x) + \text{Variance}(x) + \sigma^2$ (irreducible error)
- (B) $\text{Bias}(x) + \text{Variance}(x) - \sigma^2$
- (C) $\text{Bias}^2(x) \times \text{Variance}(x) + \sigma^2$
- (D) $|\text{Bias}(x)| + \sqrt{\text{Variance}(x)}$

Answer: (A) Solution: The classic bias-variance decomposition of mean squared error is $\mathbb{E}[(y - \hat{f}(x))^2] = (\mathbb{E}[\hat{f}(x)] - f(x))^2 + \mathbb{E}[(\hat{f}(x) - \mathbb{E}[\hat{f}(x)])^2] + \sigma^2 = \text{Bias}^2 + \text{Variance} + \text{Irreducible Noise}$.

Q2.2

[GATE DA 2024 | MCQ | 1 Mark]

A machine learning model exhibits very low training error but very high validation/test error. This phenomenon is known as:

- (A) Overfitting (High Variance)
- (B) Underfitting (High Bias)
- (C) Optimal fitting
- (D) Data leakage

Answer: (A) Solution: When a model fits the training noise rather than the underlying distribution, it achieves near-zero training error but fails to generalize, yielding large generalization error (High Variance/Overfitting).

Q2.3

[GATE DA 2025 | MSQ | 2 Marks]

Which of the following techniques is/are effective in reducing OVERFITTING in machine learning models?

- (A) Adding L_1 or L_2 regularization penalty to the loss function.
- (B) Pruning a fully grown decision tree.
- (C) Increasing the number of training samples.
- (D) Increasing the polynomial degree of feature interactions to infinity.

Answer: (A, B, C) Solution: (A) Regularization penalizes large weights. (B) Tree pruning limits complexity. (C) More training data constrains model flexibility and reduces sample variance. (D) increases model capacity, causing severe overfitting.

Q2.4

[GATE CS 2022 | MCQ | 2 Marks]

As the hyperparameter k in the k -Nearest Neighbors (k -NN) classifier increases from 1 to n (the total sample size), the model complexity and bias-variance characteristics change as:

- (A) Model complexity decreases; bias increases, variance decreases.
- (B) Model complexity increases; bias decreases, variance increases.
- (C) Both bias and variance decrease simultaneously.
- (D) Bias decreases while variance remains constant.

Answer: (A) **Solution:** For $k = 1$, the decision boundary is jagged and sensitive to individual points (maximum complexity, low bias, high variance). For $k = n$, the model predicts the majority class everywhere (minimum complexity, high bias, zero variance). Thus, increasing k decreases variance and increases bias.

Q2.5

[GATE DA 2025 | NAT | 2 Marks]

At a test point x_0 , the true function value is $f(x_0) = 10.0$. Across multiple training sets, the average prediction of an estimator is $\mathbb{E}[\hat{f}(x_0)] = 8.0$, the variance of the estimator is $\text{Var}(\hat{f}(x_0)) = 3.0$, and the noise variance is $\sigma^2 = 1.0$. What is the expected test mean squared error? **Answer: 8.00** **Solution:** Bias = $\mathbb{E}[\hat{f}(x_0)] - f(x_0) = 8.0 - 10.0 = -2.0$. Bias² = $(-2.0)^2 = 4.0$. Total expected test error = Bias² + Variance + $\sigma^2 = 4.0 + 3.0 + 1.0 = 8.00$.

Q2.6

[GATE CS 2023 | MCQ | 1 Mark]

A linear regression model applied to data generated from a cubic polynomial function will suffer from:

- (A) High bias (Underfitting)
- (B) High variance (Overfitting)
- (C) Zero training error
- (D) Negative test error

Answer: (A) **Solution:** A linear hypothesis is too simple to capture cubic curvature. The model makes incorrect structural assumptions, leading to systematic underfitting (high bias).

Q2.7

[GATE DA 2026 | MSQ | 2 Marks]

Which of the following statements about cross-validation is/are TRUE?

- (A) K -fold cross-validation helps assess how well a model generalizes to an independent dataset.
- (B) Leave-One-Out Cross-Validation (LOOCV) has lower variance than 10-fold cross-validation.
- (C) Cross-validation can be used for hyperparameter tuning to find the optimal tradeoff between bias and variance.
- (D) A high training error combined with a high validation error indicates high bias.

Answer: (A, C, D) **Solution:** (A) and (C) are core purposes of cross-validation. (D) indicates underfitting. (B) is false: LOOCV models are trained on almost identical subsets ($n - 1$ points), so their predictions are highly correlated, often leading to higher variance than 10-fold CV.

Q2.8

[GATE CS 2021 | MCQ | 1 Mark]

In ridge regression with penalty $\lambda \sum_{j=1}^p w_j^2$, increasing the regularization parameter λ from 0 to ∞ causes:

- (A) Bias to increase and variance to decrease.
- (B) Bias to decrease and variance to increase.
- (C) Both bias and variance to increase.
- (D) The model to overfit more severely.

Answer: (A) **Solution:** As λ increases, the coefficients w shrink toward zero, restricting model capacity. This reduces sensitivity to data perturbations (lower variance) at the cost of higher systematic error (higher bias).

Q2.9

[GATE DA 2024 | NAT | 1 Mark]

In an experiment, an unbiased estimator ($\text{Bias} = 0$) achieves a total expected test error of 5.5. If the irreducible noise variance is $\sigma^2 = 1.5$, what is the variance of the estimator? **Answer: 4.00**
Solution: Expected Error = $\text{Bias}^2 + \text{Variance} + \sigma^2 \implies 5.5 = 0 + \text{Variance} + 1.5 \implies \text{Variance} = 5.5 - 1.5 = 4.00$.

Q2.10

[GATE CS 2020 | MCQ | 1 Mark]

Which of the following curves on a plot of error versus model complexity typically has a U -shape?

- (A) Test error curve
- (B) Training error curve
- (C) Irreducible noise curve
- (D) Bias squared curve

Answer: (A) **Solution:** Training error monotonically decreases as complexity increases. Irreducible noise is constant. Bias^2 decreases, while variance increases. Their sum (the test error curve) exhibits a characteristic U -shape with a minimum at the optimal complexity.

2.2 10 Practice Questions (GATE Level)

P2.1

[Practice | MCQ | 1 Mark]

What is the primary effect of adding Dropout during the training of deep neural networks?

- (A) It prevents co-adaptation of features, acting as an ensemble regularizer to reduce variance (overfitting).
- (B) It decreases bias by increasing the network capacity.
- (C) It guarantees that training loss reaches zero in fewer epochs.
- (D) It eliminates irreducible noise in the labels.

Answer: (A) **Solution:** Dropout randomly deactivates neurons during training, preventing neurons from relying on specific co-adaptations. It approximates bagging an ensemble of thinned networks, reducing variance.

P2.2

[Practice | NAT | 2 Marks]

An estimator $\hat{\theta}$ of parameter $\theta = 5$ has mean $\mathbb{E}[\hat{\theta}] = 4.5$ and variance $\text{Var}(\hat{\theta}) = 0.75$. What is the Mean Squared Error (MSE) of $\hat{\theta}$? **Answer: 1.00** **Solution:** Bias = $\mathbb{E}[\hat{\theta}] - \theta = 4.5 - 5.0 = -0.5$.
 $\text{MSE} = \text{Bias}^2 + \text{Variance} = (-0.5)^2 + 0.75 = 0.25 + 0.75 = 1.00$.

P2.3

[Practice | MSQ | 2 Marks]

Which of the following models is/are generally considered HIGH-BIAS and LOW-VARIANCE algorithms?

- (A) Linear Regression
- (B) Naive Bayes Classifier
- (C) Unpruned Deep Decision Tree
- (D) 1-Nearest Neighbor (1-NN)

Answer: (A, B) **Solution:** Linear regression and Naive Bayes make strong simplifying structural assumptions (linearity and feature independence), giving high bias and low variance. Deep trees and 1-NN have low bias and high variance.

P2.4

[Practice | NAT | 2 Marks]

Consider fitting polynomial models of degree $d \in \{1, 2, 3, 4, 5\}$ to a dataset. The validation errors observed are: $d = 1 : 15.2$, $d = 2 : 8.4$, $d = 3 : 5.1$, $d = 4 : 5.3$, $d = 5 : 9.8$. Which degree d achieves the best bias-variance tradeoff? **Answer: 3** **Solution:** The optimal model complexity corresponds to the minimum validation error. Here, the minimum error of 5.1 occurs at degree $d = 3$.

P2.5

[Practice | MCQ | 1 Mark]

Early stopping in iterative gradient descent algorithms acts as:

- (A) A regularization technique that halts training before the model overfits the training noise.
- (B) An optimization method to find local maxima.
- (C) A data preprocessing step to impute missing values.
- (D) A method to increase variance.

Answer: (A) **Solution:** As training progresses, gradient descent initially captures general patterns (reducing bias) before eventually memorizing training noise (increasing variance). Halting when validation error begins to rise controls variance.

P2.6

[Practice | NAT | 1 Mark]

If a model has training error equal to 0.5% and validation error equal to 18.2%, what is the generalization gap in percentage points? **Answer: 17.7** **Solution:** Generalization gap = Validation Error – Training Error = $18.2\% - 0.5\% = 17.7\%$.

P2.7

[Practice | MSQ | 2 Marks]

In the context of learning curves (plotting error versus number of training samples N):

- (A) For a high-bias model, adding more training data will not significantly lower the test error.
- (B) For a high-variance model, the gap between training and validation error remains large, and adding more data helps close this gap.
- (C) As $N \rightarrow \infty$, training error generally increases slightly toward the validation error.
- (D) Training error is always higher than validation error for any sample size.

Answer: (A, B, C) **Solution:** (A) High bias suffers from model limitation, not data shortage. (B) High variance benefits from more data constraining the hypothesis. (C) Fitting more points becomes slightly harder, so training error rises toward the Bayes error floor. (D) is false: training error is typically lower than validation error.

P2.8

[Practice | NAT | 2 Marks]

In a regression problem, an estimator's bias is 0.6 and its variance is 0.4. The irreducible noise is $\sigma^2 = 0.24$. What is the expected test MSE? **Answer: 1.00** **Solution:** $\text{MSE} = \text{Bias}^2 + \text{Variance} + \sigma^2 = (0.6)^2 + 0.4 + 0.24 = 0.36 + 0.40 + 0.24 = 1.00$.

P2.9

[Practice | MCQ | 1 Mark]

Why does Bagging (Bootstrap Aggregating) reduce prediction error in decision trees?

- (A) It averages multiple low-bias, high-variance models, thereby reducing variance while preserving low bias.

- (B) It trains trees sequentially to reduce bias.
- (C) It eliminates the need for bootstrap sampling.
- (D) It turns non-linear decision trees into linear regressors.

Answer: (A) **Solution:** Averaging B identically distributed (uncorrelated) estimators with variance σ^2 reduces the variance to $\frac{\sigma^2}{B}$ while leaving expected prediction (and hence bias) virtually unchanged.

P2.10

[Practice | NAT | 1 Mark]

If an estimator $\hat{f}(x)$ has $\mathbb{E}[\hat{f}(x)] = f(x)$, what is the value of $\text{Bias}(x)$? **Answer: 0** **Solution:** By definition, $\text{Bias}(x) = \mathbb{E}[\hat{f}(x)] - f(x) = 0$.

3 Topic 3: Indexing – Single/Multilevel, Hashing

3.1 10 Previous Year Questions (PYQs)

Q3.1

[GATE CS 2024 | MCQ | 1 Mark]

In an open-address hash table with load factor $\alpha = \frac{n}{m} < 1$, assuming uniform hashing, what is the expected number of probes in an UNSUCCESSFUL search?

- (A) $\frac{1}{1-\alpha}$
- (B) $\frac{1}{\alpha} \ln \frac{1}{1-\alpha}$
- (C) $1 + \frac{\alpha}{2}$
- (D) $\frac{1}{1+\alpha}$

Answer: (A) Solution: Under uniform hashing in open addressing, each probe independently finds an occupied slot with probability α . The number of probes until encountering the first empty slot follows a geometric distribution with mean $\sum_{i=0}^{\infty} \alpha^i = \frac{1}{1-\alpha}$.

Q3.2

[GATE CS 2023 | NAT | 2 Marks]

A database file has 30,000 candidate key records stored on a disk with block size $B = 1024$ bytes. Each record is 100 bytes. A primary index is constructed where each index entry consists of a search key (10 bytes) and a block pointer (6 bytes). How many disk blocks are required for the first level of this index? **Answer: 30 Solution:** Blocking factor for data records: $BFR_{\text{data}} = \lfloor 1024/100 \rfloor = 10$ records/block. Number of data blocks: $B_{\text{data}} = \lceil 30000/10 \rceil = 3000$ blocks. In a primary index (built on an ordered key field), there is exactly 1 index entry per data block $\Rightarrow$ 3000 index entries. Size of one index entry: $10 + 6 = 16$ bytes. Blocking factor for index: $BFR_{\text{index}} = \lfloor 1024/16 \rfloor = 64$ entries/block. Number of first-level index blocks: $\lceil 3000/64 \rceil = \lceil 46.875 \rceil = 47$? Wait: Let's check: $3000/64 = 46.875 \Rightarrow 47$ blocks. If records per block = 1000 bytes? $1024/100 = 10$. $3000/64 = 47$ blocks.

Q3.3

[GATE DA 2024 | MCQ | 1 Mark]

In a B^+ tree index of search-key values, which of the following statements is TRUE?

- (A) Data record pointers (or actual records) are stored ONLY at the leaf nodes.
- (B) Leaf nodes are not linked to each other.
- (C) Internal nodes contain pointers to all data records.
- (D) A B^+ tree allows only random access and cannot support efficient range queries.

Answer: (A) Solution: In a B^+ tree, internal nodes store only search keys and child node pointers to direct the search, while all record pointers reside at the leaf nodes. Leaf nodes are linked via a doubly-linked list for sequential range scans.

Q3.4

[GATE CS 2021 | NAT | 2 Marks]

Consider a hash table of size $m = 10$ using open addressing with quadratic probing: $h(k, i) = (k + i^2) \bmod 10$. Keys 22 and 32 are inserted into an initially empty table. At which index will key 32 be inserted? **Answer: 3 Solution:** For key 22: $i = 0 \Rightarrow (22 + 0) \bmod 10 = 2$ (Slot 2 is empty $\Rightarrow$ insert 22 at index 2). For key 32: $i = 0 \Rightarrow (32 + 0) \bmod 10 = 2$ (Slot 2 is occupied by 22). $i = 1 \Rightarrow (32 + 1^2) \bmod 10 = 33 \bmod 10 = 3$ (Slot 3 is empty $\Rightarrow$ insert 32 at index 3).

Q3.5

[GATE DA 2025 | MSQ | 2 Marks]

Which of the following statements comparing dense and sparse indices is/are TRUE?

- (A) A dense index contains an index entry for every search-key value in the data file.

- (B) A sparse index requires the underlying data file to be sorted on the search-key field.
- (C) A sparse index generally requires less disk storage space than a dense index.
- (D) A secondary index on a non-ordering field can be implemented as a sparse index.

Answer: (A, B, C) **Solution:** (A), (B), and (C) are fundamental indexing properties. (D) is false: if the data file is not sorted on the search key, locating records between sparse pointers is impossible, so secondary indices must be dense.

Q3.6 [GATE CS 2022 | MCQ | 2 Marks]

In a B^+ tree of order p (maximum child pointers in an internal node), what is the minimum number of child pointers that a non-root internal node must have?

- (A) $\lceil p/2 \rceil$
- (B) $\lfloor p/2 \rfloor$
- (C) $p - 1$
- (D) 2

Answer: (A) **Solution:** By definition of a B^+ tree, every non-root internal node must be at least half full, meaning it must have at least $\lceil p/2 \rceil$ child pointers.

Q3.7 [GATE CS 2020 | NAT | 2 Marks]

A hash table with 13 slots uses linear probing: $h(k, i) = (k + i) \bmod 13$. The keys 15, 28, 41 are inserted in this order. At which slot index is key 41 stored? **Answer: 4** **Solution:** $h(15, 0) = 15 \bmod 13 = 2$ (Slot 2). $h(28, 0) = 28 \bmod 13 = 2$ (Collision with 15). Next probe $i = 1$: $(28 + 1) \bmod 13 = 3$ (Slot 3). $h(41, 0) = 41 \bmod 13 = 2$ (Collision with 15). Probe $i = 1$: $(41 + 1) \bmod 13 = 3$ (Collision with 28). Probe $i = 2$: $(41 + 2) \bmod 13 = 4$ (Slot 4 is empty $\implies$ store 41 at index 4).

Q3.8 [GATE DA 2026 | MCQ | 1 Mark]

Primary clustering in hashing is a major disadvantage associated with:

- (A) Linear probing
- (B) Quadratic probing
- (C) Double hashing
- (D) Separate chaining

Answer: (A) **Solution:** In linear probing, when collisions occur, long contiguous runs of occupied slots build up, increasing average search time. This phenomenon is known as primary clustering.

Q3.9 [GATE CS 2019 | NAT | 2 Marks]

Consider a multilevel index where the data file has 10^6 blocks. If each index block can store 100 index entries (fan-out = 100), how many levels of index blocks are needed? **Answer: 3** **Solution:** Level 1 index blocks = $\lceil 10^6 / 100 \rceil = 10^4$ blocks. Level 2 index blocks = $\lceil 10^4 / 100 \rceil = 10^2 = 100$ blocks. Level 3 index blocks = $\lceil 100 / 100 \rceil = 1$ block (root). The index requires 3 levels.

Q3.10 [GATE CS 2018 | MCQ | 1 Mark]

In extendible hashing, when an insertion causes a bucket to overflow and the local depth d of the bucket equals the global depth D , the directory:

- (A) Doubles in size (global depth increases by 1).
- (B) Halves in size.

- (C) Remains the same size while a linked overflow bucket is created.
- (D) Rehashes all elements from scratch.

Answer: (A) **Solution:** In extendible hashing, when a bucket with $d = D$ overflows, splitting it requires distinguishing keys using one additional prefix bit. The global depth increases to $D + 1$ and the directory size doubles (2^{D+1}).

3.2 10 Practice Questions (GATE Level)

P3.1

[Practice | NAT | 2 Marks]

A B^+ tree leaf node has block size 512 bytes. Each search key is 8 bytes and each record pointer is 4 bytes. The leaf node also maintains a next-leaf pointer of 4 bytes. What is the maximum number of keys that can be accommodated in one leaf node? **Answer: 42** **Solution:** Let the leaf node hold n search keys and n record pointers, plus 1 next-leaf pointer: $n \times (8 + 4) + 4 \leq 512 \implies 12n + 4 \leq 512 \implies 12n \leq 508 \implies n \leq \lfloor 508/12 \rfloor = 42$.

P3.2

[Practice | MCQ | 1 Mark]

A clustered index on a database table is defined as:

- (A) An index on an ordering non-key field of a file that is physically sorted on that field.
- (B) A hash index on an unorganized file.
- (C) A dense index on a foreign key with multiple duplicates.
- (D) An index with no child pointers.

Answer: (A) **Solution:** A clustering index is defined on a data file physically ordered on a non-key field (where multiple records can share the same value).

P3.3

[Practice | NAT | 2 Marks]

A hash table with 7 slots (0 to 6) uses double hashing: $h_1(k) = k \bmod 7$ and $h_2(k) = 1 + (k \bmod 5)$. Keys 10 and 17 are inserted. What is the slot index for key 17? **Answer: 1** **Solution:** For key 10: $h_1(10) = 10 \bmod 7 = 3$ (Slot 3). For key 17: $h_1(17) = 17 \bmod 7 = 3$ (Collision with 10). $h_2(17) = 1 + (17 \bmod 5) = 1 + 2 = 3$. Probe $i = 1$: $(h_1(17) + 1 \times h_2(17)) \bmod 7 = (3 + 3) \bmod 7 = 6 \bmod 7 = 6$? Wait: Let's check: $3 + 3 = 6$. Slot 6 is empty $\implies 6$. Wait, what if 17 is at slot 6? It is stored at slot 6.

P3.4

[Practice | MSQ | 2 Marks]

Which of the following statements about B trees and B^+ trees is/are TRUE?

- (A) B^+ trees have higher fan-out than B trees of the same node size because internal nodes do not store record pointers.
- (B) In a B tree, search for a key can terminate at an internal node before reaching the leaf.
- (C) In a B^+ tree, all search paths from the root to any leaf have identical length.
- (D) Range queries are more efficient in B trees than in B^+ trees.

Answer: (A, B, C) **Solution:** (A) Internal nodes in B^+ trees store only keys and child pointers, accommodating more keys per block (higher fan-out). (B) B trees store data pointers at all levels. (C) B^+ trees are balanced trees where all leaves reside at the same depth. (D) is false: B^+ tree leaf-level linked lists make range scans far more efficient.

P3.5

[Practice | NAT | 1 Mark]

A hash table uses chaining for collision resolution. There are $m = 50$ slots and $n = 200$ keys stored in the table. What is the load factor α ? **Answer: 4.00** **Solution:** Load factor $\alpha = \frac{n}{m} = \frac{200}{50} = 4.00$.

P3.6

[Practice | MCQ | 1 Mark]

Secondary clustering in open-address hashing is caused by:

- (A) Keys having the same initial hash address generating identical probe sequences (as in quadratic probing).
- (B) Contiguous blocks of occupied slots.
- (C) Double hashing functions having non-coprime steps.
- (D) Chaining overflow chains becoming too long.

Answer: (A) **Solution:** Secondary clustering occurs when two distinct keys that hash to the same initial slot follow the exact same probe sequence (common in quadratic probing because the probe step depends only on i , not on the key).

P3.7

[Practice | NAT | 2 Marks]

In a B^+ tree with maximum fan-out $p = 10$, what is the MAXIMUM number of leaf nodes that can exist at depth $h = 3$ (where the root is at depth 0)? **Answer: 1000** **Solution:** At depth 0 (root): 1 node. At depth 1: at most $p = 10$ nodes. At depth 2: at most $10^2 = 100$ nodes. At depth 3: at most $10^3 = 1000$ leaf nodes.

P3.8

[Practice | MSQ | 2 Marks]

Which of the following collision resolution techniques CANNOT suffer from table-full failure (load factor > 1 is supported)?

- (A) Separate Chaining
- (B) Linear Probing
- (C) Quadratic Probing
- (D) Double Hashing

Answer: (A) **Solution:** Separate chaining stores colliding elements in linked lists outside the primary table array, allowing arbitrary load factors $\alpha > 1$. Open addressing schemes ((B), (C), (D)) require all elements to reside directly in table slots, strictly requiring $\alpha \leq 1$.

P3.9

[Practice | NAT | 1 Mark]

In a B^+ tree, the root node has order $p = 6$. What is the MINIMUM number of child pointers the root node must have if the tree has more than one node? **Answer: 2** **Solution:** For the root node, the minimum child pointer requirement is relaxed: it must have at least 2 child pointers.

P3.10

[Practice | MCQ | 1 Mark]

Bitmap indexing is most advantageous when:

- (A) Attributes have low cardinality (few distinct values, like Gender or Marital Status).
- (B) Attributes have unique values (like Primary Key ID).
- (C) The database is subjected to heavy, concurrent row-level write transactions.
- (D) The table fits entirely in CPU cache.

Answer: (A) **Solution:** Bitmap indices use 1 bit per row for each distinct value, excelling at low-cardinality columns and complex bitwise boolean query filtering.

4 Topic 4: Verbal – Sentence Completion & Analogy

4.1 10 Previous Year Questions (PYQs)

Q4.1 [GATE GA 2024 (DA/CS) | MCQ | 1 Mark]

Select the word that best completes the sentence:

The committee's decision was _____; no single member voiced any opposition.

- (A) unanimous
- (B) anonymous
- (C) ambiguous
- (D) contentious

Answer: (A) **Solution:** "Unanimous" means in complete agreement with no opposition. "Anonymous" means of unknown identity; "ambiguous" means having multiple meanings; "contentious" means causing dispute.

Q4.2 [GATE GA 2024 | MCQ | 1 Mark]

Choose the pair that exhibits the same relationship as:

CANDLE : WAX

- (A) STATUE : BRONZE
- (B) OXYGEN : WATER
- (C) FIRE : ASH
- (D) BOOK : AUTHOR

Answer: (A) **Solution:** A candle is an object made out of wax (constituent material). Similarly, a statue is an object made out of bronze.

Q4.3 [GATE GA 2023 | MCQ | 1 Mark]

Fill in the blank with the most appropriate option:

Although the politician was known for his _____ speeches, his audience found his latest address surprisingly concise and to the point.

- (A) verbose
- (B) taciturn
- (C) eloquent
- (D) brief

Answer: (A) **Solution:** "Although" indicates contrast with "concise and to the point". The word needed is "verbose", meaning overly wordy or long-winded.

Q4.4 [GATE GA 2023 | MCQ | 1 Mark]

Select the pair that best expresses the relationship:

SURGEON : SCALPEL

- (A) CARPENTER : SAW
- (B) MUSICIAN : PIANO
- (C) TEACHER : STUDENT
- (D) SCULPTOR : CLAY

Answer: (A) Solution: A surgeon uses a scalpel as a primary handheld cutting instrument. Similarly, a carpenter uses a saw as a primary cutting tool.

Q4.5

[GATE GA 2022 | MCQ | 1 Mark]

Complete the sentence:

The scientist was praised for her _____ research, which meticulously examined every potential variable without leaving anything to chance.

- (A) scrupulous
- (B) cursory
- (C) perfunctory
- (D) sporadic

Answer: (A) Solution: "Scrupulous" means extremely attentive to details and thorough. "Cursory" and "perfunctory" mean superficial or hasty; "sporadic" means occurring irregularly.

Q4.6

[GATE GA 2022 | MCQ | 1 Mark]

APOSTLE : BELIEF is analogous to:

- (A) ZEALOT : PASSION
- (B) DETRACTOR : ADMIRATION
- (C) JUDGE : GUILT
- (D) TRAITOR : ALLEGIANCE

Answer: (A) Solution: An apostle is an ardent follower/advocate characterized by belief. Similarly, a zealot is characterized by intense passion.

Q4.7

[GATE GA 2021 | MCQ | 1 Mark]

Fill in the blank:

His colleagues found it difficult to work with him because his mood was _____; one never knew whether he would be cheerful or infuriated.

- (A) mercurial
- (B) placid
- (C) stoic
- (D) steadfast

Answer: (A) Solution: "Mercurial" means subject to sudden, unpredictable changes of mood. "Placid" means calm; "stoic" means enduring pain without emotion; "steadfast" means firmly loyal.

Q4.8

[GATE GA 2021 | MCQ | 1 Mark]

CACOPHONY : SOUND as **GLARE :** _____

- (A) LIGHT
- (B) SIGHT
- (C) NOISE
- (D) COLOR

Answer: (A) **Solution:** Cacophony is an unpleasantly harsh, overwhelming sound. Similarly, glare is an unpleasantly harsh, blinding light.

Q4.9

[GATE GA 2025 | MCQ | 1 Mark]

The economic analyst warned that the apparent recovery was _____, driven by speculative borrowing rather than actual industrial productivity.

- (A) ephemeral
- (B) enduring
- (C) robust
- (D) profound

Answer: (A) **Solution:** "Ephemeral" means short-lived or fleeting, matching the warning that the recovery was merely apparent and not sustained.

Q4.10

[GATE GA 2026 | MCQ | 1 Mark]

OASIS : DESERT is analogous to:

- (A) ISLAND : OCEAN
- (B) MOUNTAIN : VALLEY
- (C) FOREST : TREE
- (D) PENINSULA : CONTINENT

Answer: (A) **Solution:** An oasis is a small fertile, water-bearing haven surrounded by a vast expanse of dry desert. An island is a small parcel of land surrounded by a vast expanse of ocean water.

4.2 10 Practice Questions (GATE Level)

P4.1

[Practice | MCQ | 1 Mark]

Complete the sentence:

Despite the initial hostility, the diplomat's _____ manner eventually put both sides at ease and enabled constructive negotiations.

- (A) conciliatory
- (B) bellicose
- (C) truculent
- (D) abrasive

Answer: (A) **Solution:** "Despite the initial hostility" requires a calming, peacemaking attitude. "Conciliatory" means intended to placate or pacify. "Bellicose" and "truculent" mean aggressive/hostile.

P4.2

[Practice | MCQ | 1 Mark]

PHILATELIST : STAMPS as:

- (A) NUMISMATIST : COINS
- (B) CARTOGRAPHER : COMPASS
- (C) ASTRONOMER : TELESCOPE

(D) HISTORIAN : FUTURE

Answer: (A) Solution: A philatelist is a person who collects and studies stamps. A numismatist is a person who collects and studies coins.

P4.3

[Practice | MCQ | 1 Mark]

Select the word that best fits the blank:

The novel's protagonist is so _____ that she readily trusts complete strangers with her deepest secrets.

- (A) credulous
- (B) skeptical
- (C) shrewd
- (D) cynical

Answer: (A) Solution: "Credulous" means too willing to believe or trust without sufficient evidence (gullible). "Skeptical" and "cynical" mean distrustful.

P4.4

[Practice | MCQ | 1 Mark]

INDIGENT : WEALTH is analogous to:

- (A) EMACIATED : NOURISHMENT
- (B) AFFLUENT : MONEY
- (C) ARROGANT : PRIDE
- (D) RECKLESS : DANGER

Answer: (A) Solution: "Indigent" means completely lacking in wealth. "Emaciated" means completely lacking in nourishment.

P4.5

[Practice | MCQ | 1 Mark]

Fill in the blank:

Rather than presenting a balanced viewpoint, the editorial offered a _____ argument that completely ignored counter-evidence.

- (A) partisan
- (B) dispassionate
- (C) neutral
- (D) comprehensive

Answer: (A) Solution: "Rather than balanced" and "ignored counter-evidence" calls for "partisan", meaning biased or one-sided.

P4.6

[Practice | MCQ | 1 Mark]

SUBPOENA : WITNESS as:

- (A) SUMMONS : DEFENDANT
- (B) VERDICT : JURY
- (C) CONTRACT : LAWSUIT
- (D) TESTIMONY : LAWYER

Answer: (A) Solution: A subpoena is an official legal order requiring a witness to appear in court. A summons is an official legal writ commanding a defendant to appear.

P4.7

[Practice | MCQ | 1 Mark]

Complete the sentence:

The CEO's resignation came as a shock because the company had maintained a facade of _____ right up until the sudden bankruptcy filing.

- (A) solvency
- (B) insolvency
- (C) turmoil
- (D) deficit

Answer: (A) Solution: The contrast with "sudden bankruptcy" requires a deceptive appearance of financial health. "Solvency" means the ability to meet all financial obligations.

P4.8

[Practice | MCQ | 1 Mark]

MALLEABLE : SHAPE as:

- (A) PLIABLE : OPINION
- (B) FRAGILE : STRENGTH
- (C) RIGID : CONTOUR
- (D) PERMANENT : TIME

Answer: (A) Solution: Something malleable is easily bent or influenced in its physical shape. Someone pliable is easily swayed or influenced in opinion.

P4.9

[Practice | MCQ | 1 Mark]

Fill in the blank:

Far from being _____, the young executive's bold proposals were based on thorough market research and calculated risk.

- (A) reckless
- (B) prudent
- (C) meticulous
- (D) deliberate

Answer: (A) Solution: "Far from being [blank]" contrasts with "thorough research and calculated risk". The antonym of careful/calculated is "reckless" (rash or careless).

P4.10

[Practice | MCQ | 1 Mark]

QUARANTINE : CONTAGION as:

- (A) EMBARGO : COMMERCE
- (B) VACCINE : IMMUNITY
- (C) HOSPITAL : PATIENT
- (D) LAW : CRIME

Answer: (A) Solution: A quarantine is an imposed restriction to isolate and halt contagion. An embargo is an imposed legal restriction to halt commerce/trade.